personal portfolio

Muhammad Hafizh Rifandi
Data Analyst & Business Intelligence





About Hafizh Rifandi

Fresh Graduated from Universiti Teknologi Malaysia with Bachelor Degree Mechanical Engineering - Aeronautics (Hons) with hands-on experience across engineering, technical operations, maintenance, and data-driven problem solving.

Currently transitioning into Data Analytics and Business Intelligence, combining an engineering background with analytical and technical skills to transform data into actionable insights and support business decision-making.

Software :

Excel Python MySQL Power BI Tableau

Education

My educational background combines science, mechanical engineering, and data analytics, building a strong foundation in analytical thinking, technical problem-solving, and data-driven decision making.



Nurul Fikri Boarding School Lembang

Science Class



Universiti Teknologi Malaysia

Bachelor of Mechanical Engineering - Aeronautics (With Honours)



Dibimbing Bootcamp

Data Analyst & Business Intelligence

Work Experience



Pertamina EP KSO - Samudra Energy BWP Meruap

Mechanical Student Internship

Supported oil and gas operations through mechanical troubleshooting and production processes, including equipment monitoring, mechanical troubleshooting, and operational activities



PT. Indopelita Aircraft Services

Aeronautical Intership

Supported aircraft maintenance activities by reviewing Aircraft Maintenance Manuals, observing turbine engine operations, and assisting with maintenance preparation for ATR-72 wing panels

Work Experience



PT. Nam Air

Technical Support Internship

Supported aircraft design and technical evaluation activities, including mechanical drawing reviews and analysis of Airworthiness Directives (AD) and Service Bulletins (SB) for the ATR-72 fleet



Xin Network Sdn Bhd

Freelance Data Infrastructure Engineer

Supported data center infrastructure activities, including server and network equipment installation, structured fiber optic, hardware inspection, and asset tagging

Main Project

Project Background

Automotive After-Sales Business Intelligence

This project analyzes automotive after-sales service operations using Work Order transaction data and Master Parts data to evaluate service performance, spare-part demand, transaction value, and customer engagement.

The project applies channel segmentation to compare Work Order patterns across different service and engagement channels, focusing on service demand, booking and promotion activity, transaction value, spare-part contribution, and operational performance

Tools : Python, Google Colab, Pandas, Excel, Power BI, DAX
Project Type : End-to-End Business Intelligence
Dashboard : Power BI Dashboard
Dataset : After Sales Dataset - Service Performance, Spare-Part Demand, and Customer Engagement Analysis
Google Colab : After Sales Analysis

Executive Summary

This project transforms automotive after-sales service and spare-part data into actionable business insights. Using Python and Google Colab, I performed data understanding, quality assessment, preprocessing, feature engineering, exploratory analysis, and business analysis. The processed data was then modeled and visualized in Power BI to monitor service performance, spare-part contribution, and customer engagement.

Raw Work Order Records

120,737

Rows

Combined from two sheets

Master Parts Records

61,970

Rows

Spare Parts Reference

Analytical Work Orders

49,136

Work Orders

Across the available period

Power BI Model

3 Tables

Fact and Dimension

Raw Dataset Overview

Work Order	Master Parts
Work Order : woid, no_work_order	Part Number, Deskripsi1, Kelompok Barang
Vehicle : pittype, kmmotor, tahun_rakit, type_motor	-
Customer : alamatdatang, kota, tanggalpembelian, pekerjaanpemilik, custjobname	-
Service: jenis_service, servicecategory, pittype	-
Sparepart: partid, description, total_perpart	-
Financial : total_njnb, total_nsc, total_perjasa	-
Operational timestamps : waktu_start_pekerjaan, waktu_finish_pekerjaan, waktu_finish_final_check, waktu_finish_pembayaran	-
Engagement/promotion : description.1, promoid, promoname, source, bookingnumber, resultpromosi, dll.	-

	Sheet 1		Sheet 2				
Work Order (Raw)	58,649 Rows	38 Columns	62,088 Rows	38 Columns	**Master Parts (Raw)**	61,970 Rows	3 Columns

Objectives

Develop a data-driven BI solution to evaluate service operations, spare-part contribution, and engagement patterns to support business decision-making.

01

Service Performance

Analyze service demand and operational duration

02

Spare-Part Demand

Identify part utilization and demand patterns

03

Customer Engagement

Analyze channels, promotion , booking and channel segmentation

04

BI Dashboard & Recommendations

Monitor key performance indicators &Turn findings into business recommendations

Problem Statement

The available after-sales data contains information across service operations, spare-part usage, and customer engagement. However, limited visibility into service performance, spare-part contribution, and channel-level engagement makes it difficult to identify meaningful business patterns and support data-driven decision-making.



Service Performance

Limited visibility into service demand, duration, and service mix makes it difficult to identify operational patterns and areas for improvement.



Spare-Part Management

Limited visibility into spare-part utilization, value contribution, and demand makes it difficult to identify the most important parts.



Customer Engagement

Fragmented booking, promotion, and channel records make it difficult to compare engagement patterns and transaction value.

Project Flow

01

Data Understanding

- Dataset Overview
- Data Structure
- Column & Metrics Understanding
- Initial Data Quality Check

02

Business Understanding

- Business Problem
- Project Goals
- Business Question

03

Data Cleaning & Preprocessing

- Handling Missing Value
- Duplicate Treatment
- Standardization
- Technical Feature Engineering
- Business Feature Engineering
- Handling Outlier

04

EDA

- Descriptive Statistics
- Distribution Analysis
- Trend Analysis
- Relationship Analysis

05

Business Insights

- Key Findings
- Recommendations

06

Dashboard

- Transforming raw data into visualization using Power BI

Project Goals

Transform after-sales service data into actionable business insights for operational and service performance improvement and Analyze channel segmentation.

Service Demand Analysis

Understand after-sales service demand across service categories, vehicle types, and locations to identify major service patterns and user demands

Operational Performance

Examine which are service activities record the highest average operational duration?

Promotion & Booking Analysis

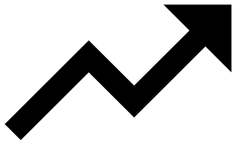
Analyze promotion and booking activities across service channels to understand their association with after-sales service activity.

Spare-Part Analysis

Identify the most utilized spare parts and part groups and evaluate their contribution to spare-part value.

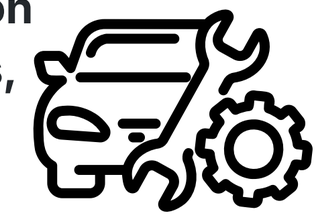
Business Questions

How are the main patterns in after-sales service demand across service categories, vehicle types, and locations?



- What drives after-sales service activity?

How does operational duration vary across service activities, and which activities have the longest average duration?



- Which service activities record the highest average operational duration?

How are promotions and booking activities associated with after-sales service activity?



- How are promotion and booking activities distributed across channels?

Which spare parts and part groups contribute most to spare-part utilization and value?



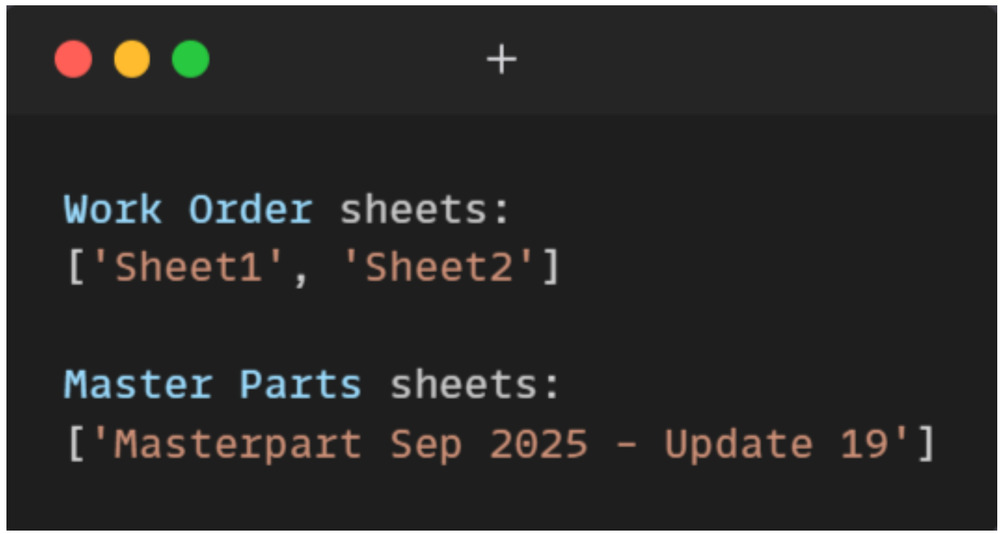
- Which parts and part groups are most utilized?

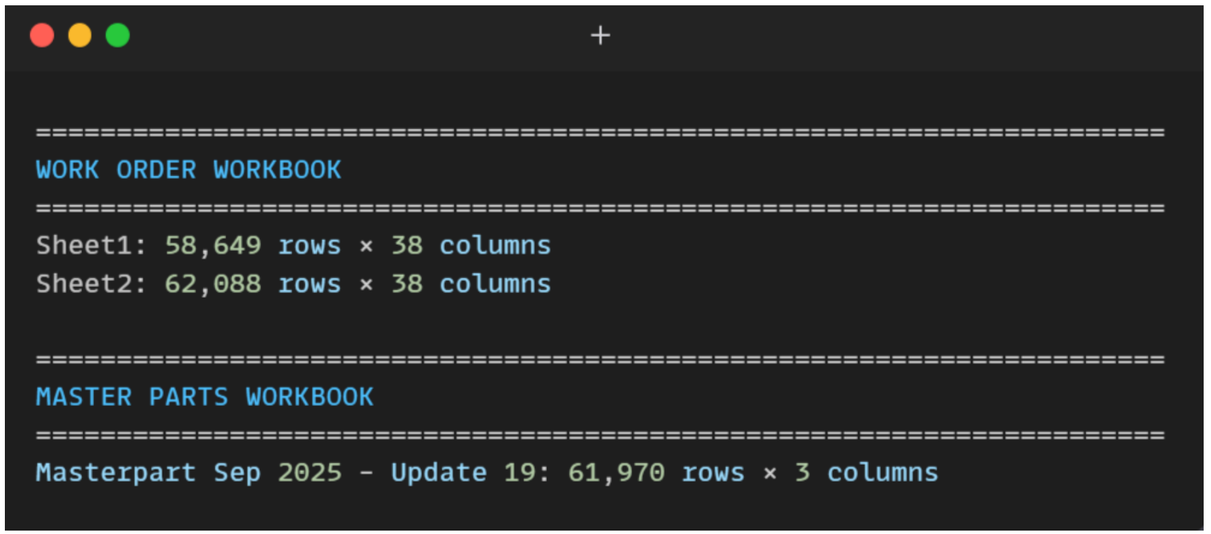
Raw vs Cleaned Data

Aspect	Raw Data	Clean Data
Work Order records	120,737	120,737
Work Order columns	38	35
Master Parts records	61,970	61,970
Date/Time format	Mixed object / datetime	Standardized datetime64[ns]
Identifier format	Mixed object/string	Standardized string
Numeric fields	Mixed numeric dtypes	Standardized numeric dtypes
Categorical values	Formatting variations	Standardized whitespace/case where required
Irrelevant columns	3 high-missing columns	Removed

1 Dataset Inventory

This project uses two primary data sources: Work Order transaction data and Master Parts reference data. **Work Order** dataset serves as the primary source of service transaction data, while the **Master Parts** dataset provides reference information to enrich spare-part descriptions and classifications.





2 Data Understanding & Preprocessing

Improving data consistency while preserving valid transactional records.

Raw Work Order Records

120,737

Rows

Combined from two sheets

Source Sheets

2

Combined Dataset

Columns Removed

3

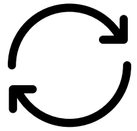
Selected Columns

Key Findings :



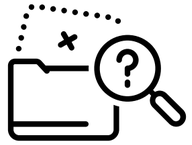
Inconsistent data formats

Standardized date formats and data types for consistent analysis.



Repeated records

16,528 exact duplicate rows identified across both sheets.



Missing values

Three columns with high missingness and limited analytical value were removed; other missing values were retained.

2 Data Understanding & Preprocessing

Model Name Standardization :

Normalized model names with multiple original variants:

type_motor_normalized	number_of_variants
BEAT SPORTY CBS ISS	2
BEAT SPORTY CBS	2
BEAT SPORTY CBS ISS DELUXE	2
GENIO CBS	2
GENIO CBS ISS	2
BEAT STREET	2
BEAT SPORTY DLX SMART KEY	2
NEW SUPRA GTR150 SPORTY	2
NEW VARIO 150	2
NEW SUPRA GTR150 EXCLUSIVE	2

Total normalized model groups with multiple variants: 10

Standarized Datetime:

	data_type
tanggalpembelian	datetime64[ns]
tanggal_input_wo	datetime64[ns]
tgl_service	datetime64[ns]
waktu_start_pekerjaan	datetime64[ns]
waktu_finish_pekerjaan	datetime64[ns]
waktu_finish_final_check	datetime64[ns]
waktu_finish_pembayaran	datetime64[ns]
paymentdate	datetime64[ns]

Missing Value :

	dataset	sheet	total_rows	duplicate_rows	duplicate_percentage	unique_rows
0	Work Order	Sheet1	58649	8119	13.84	50530
1	Work Order	Sheet2	62088	8409	13.54	53679
2	Master Parts	Masterpart Sep 2025 - Update 19	61970	0	0.00	61970

Dropped Columns :

```
Dropped columns:
['markettratementcode', 'isdropandgo', 'resultdisclaimer']
```

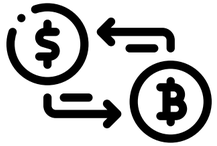
3 Feature Engineering



Time-Based Features

Service year, month, month name, day, and day name.

Purpose: Analyze service patterns over time.



Transaction Metrics

Service value, spare-part value, and total transaction value.

Purpose: Measure business contribution.



Operational Features

Operational duration derived from available work timestamps.

Purpose: Examine service duration.



Engagement Features

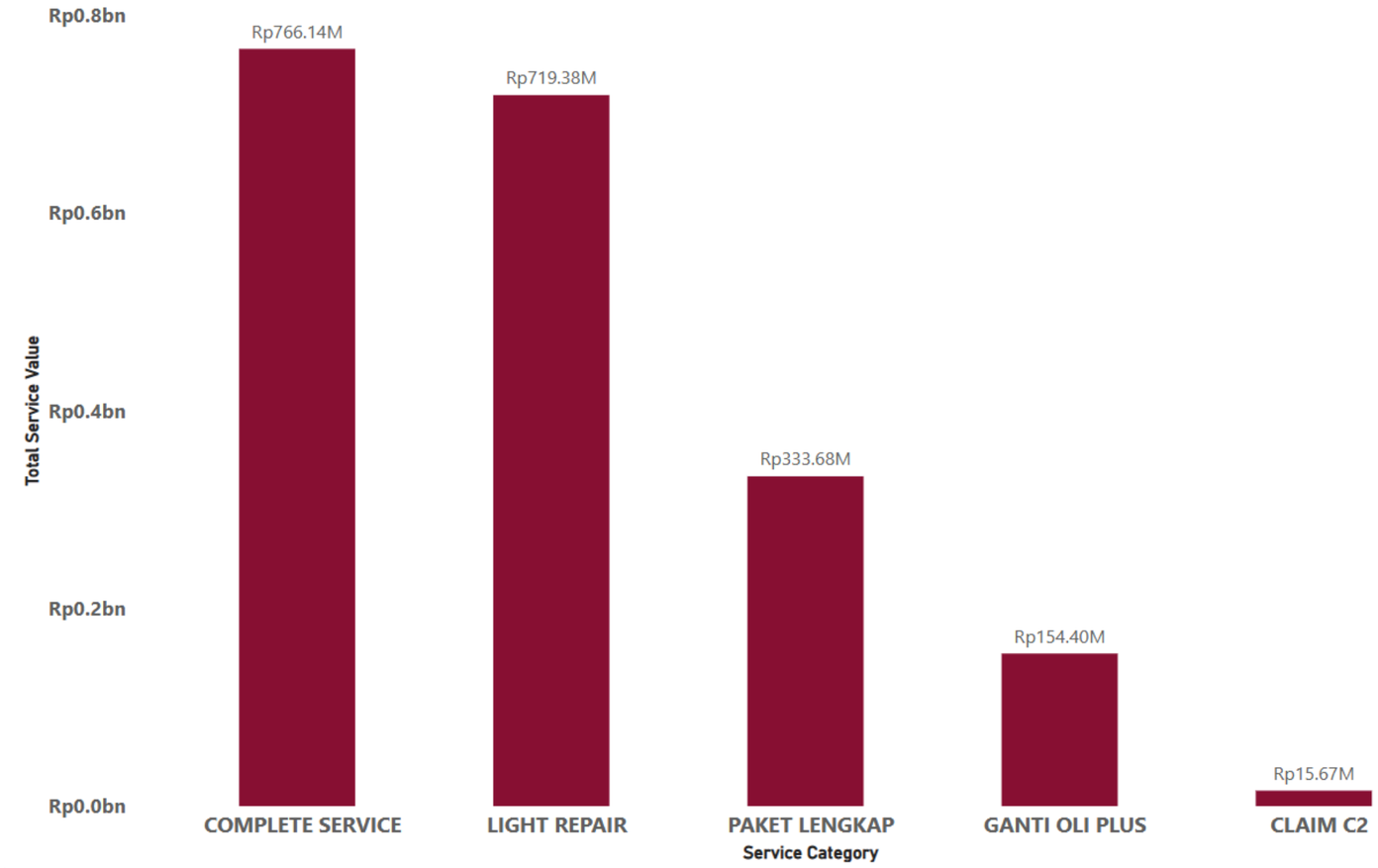
Promotion flags, booking flags, and engagement segments.

Purpose: Compare engagement patterns.

4 Exploratory Data Analysis

01 - Service Category Analysis

Total Service Value by Service Category.



- **COMPLETE SERVICE** generates the highest service value at Rp766.14M, slightly exceeding **LIGHT REPAIR** at Rp719.38M. Together, these two categories account for the largest share of total service value, while CLAIM C2 contributes the least at Rp15.67M.

Top Service

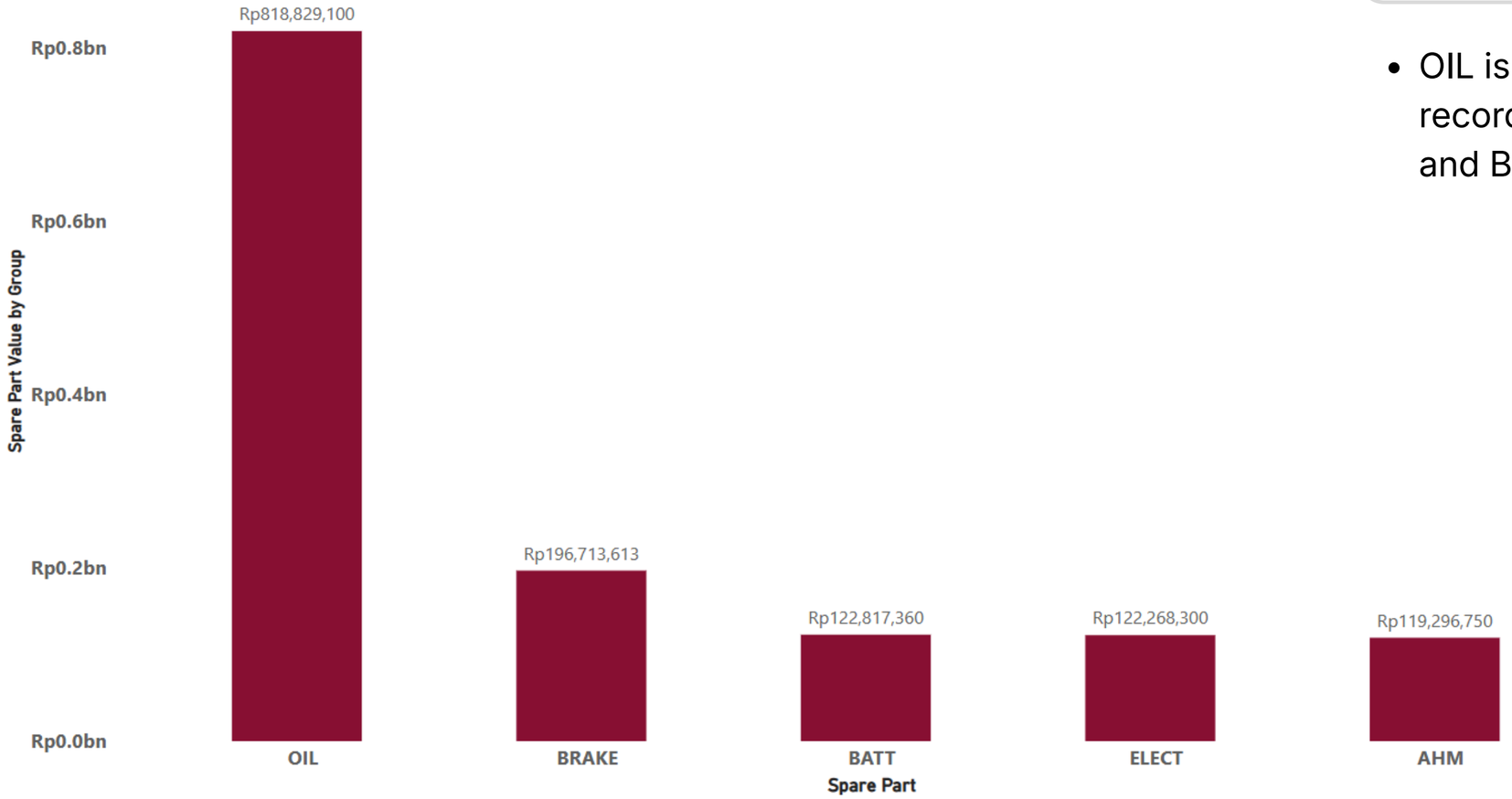
COMPLETE SERVICE

Rp. 766.14 M

4 Exploratory Data Analysis

02 - Spare Part Contribution

Total Spare-part Value by Spare Part



Top Parts Group

OIL

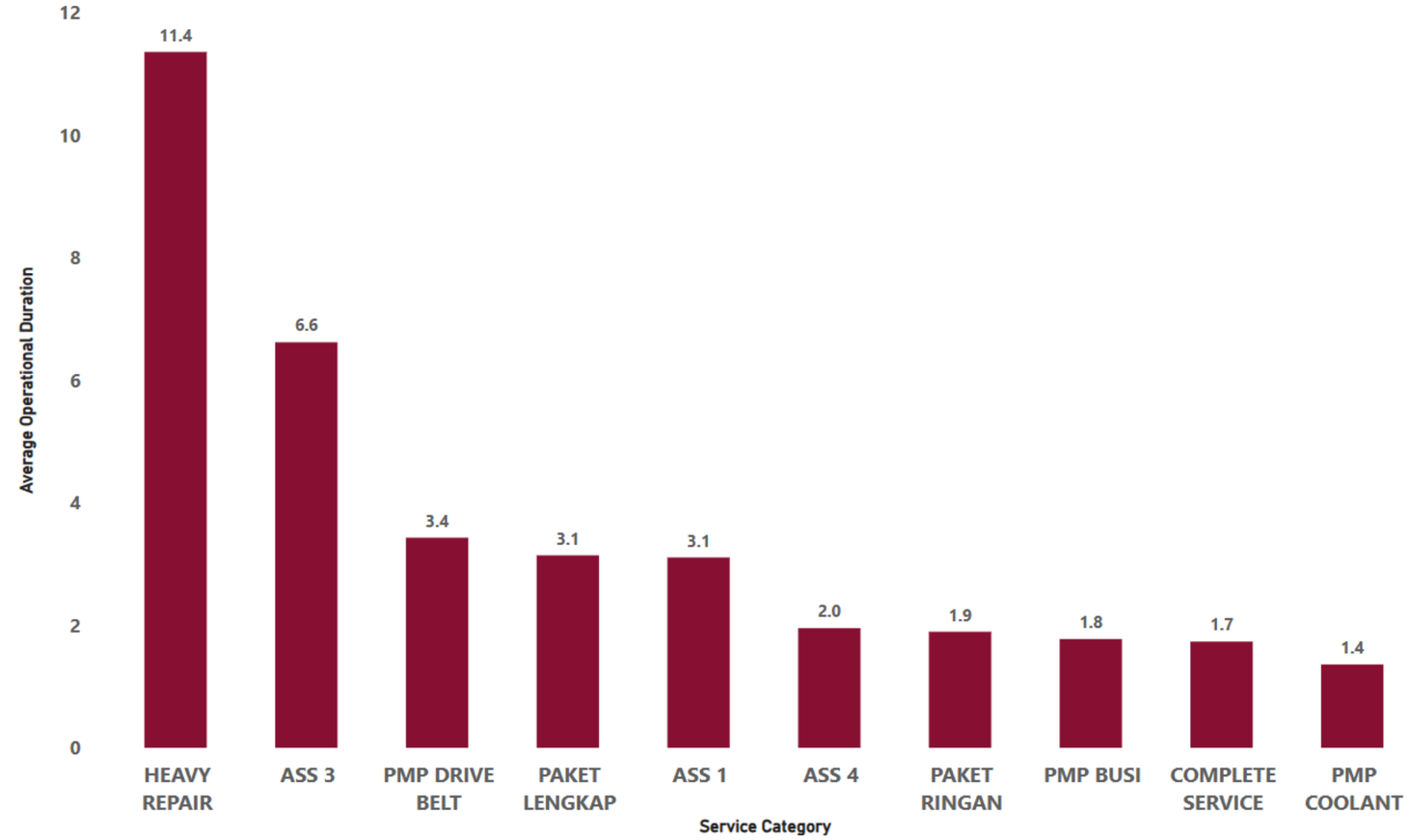
Rp. 818.8 M

- OIL is the largest spare-part group by recorded value, followed by BRAKE and BATT.

4 Exploratory Data Analysis

03 - Average Operational Duration

Average Operational Duration by Service Category



Highest Average Operational Duration

Heavy Repair

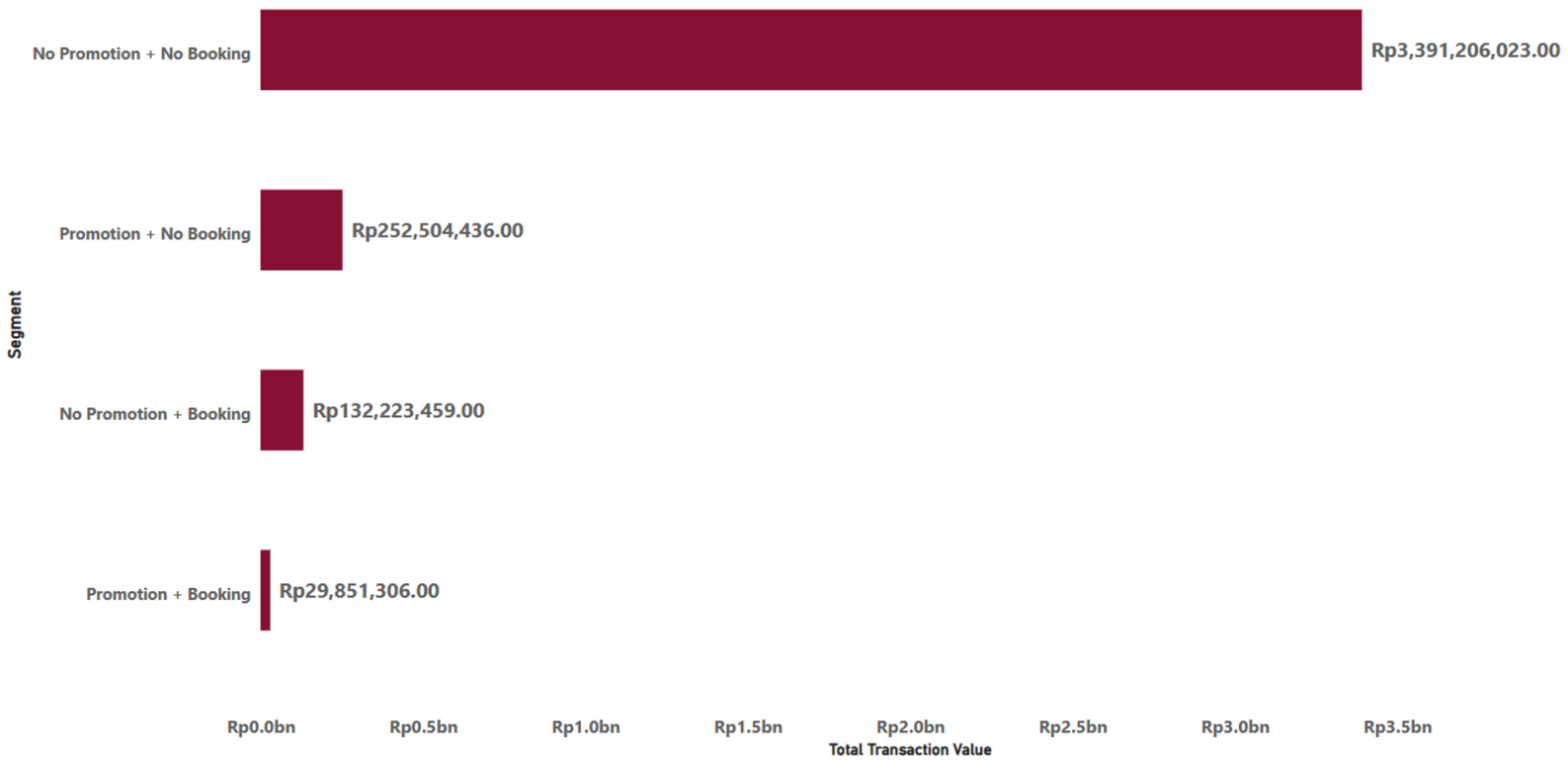
11.4 Hours

- **HEAVY REPAIR** records the longest average operational duration at 11.4, followed by **ASS 3** at 6.6. This indicates that **HEAVY REPAIR** requires substantially more operational time than other service activities, while **COMPLETE SERVICE** averages 1.7, reflecting a shorter operational duration.

4 Exploratory Data Analysis

04 - Customer Engagement Analysis

Total Transaction Value by Engagement Segment.



- **No Promotion + No Booking** segment contributes the highest transaction value (Rp3.39 billion). Promotion-associated segments contribute substantially less overall, reflecting their smaller share of recorded Work Orders.

Total Transaction Value	No Promotion + No Booking
Rp3.81B	**Rp3.39B**

5 Business Insight Summary

Volume Service Categories

- **LIGHT REPAIR** leads the highest service volume (20,377 Work Orders). Monitor service demand and resource allocation while evaluating the higher average transaction value of **COMPLETE SERVICE**.

Contribution Spare Parts

- **OIL**, **BRAKE**, and **BATT** are the leading spare-part groups by recorded value. Prioritize demand monitoring and inventory reviews for these categories to support replenishment planning.

5 Business Insight Summary



Promotion and Booking Performance

- Promotion-associated transactions show a higher average transaction value in the descriptive analysis. Compare transaction value and Work Order volume across engagement segments before expanding campaigns.



Channel Recording

- **MOBILESOBI** is the most frequently recorded service channel, accounting for **76.08%** of Work Orders with a recorded channel. Improving channel data completeness would help provide a clearer picture of service channel performance.

6 Business Recommendations

- **Optimize Service & Spare-Part Performance**



Monitor service volume and value separately

- Track Work Order volume, average transaction value, and service revenue by category to support service planning and performance reviews.



Prioritize high-contribution spare-part groups

- Review demand and recorded value for **OIL**, **BRAKE**, and **BATT** to inform inventory monitoring and replenishment planning.

6 Business Recommendations

- **Improve Engagement & Data Reliability**



Evaluate promotion and booking performance

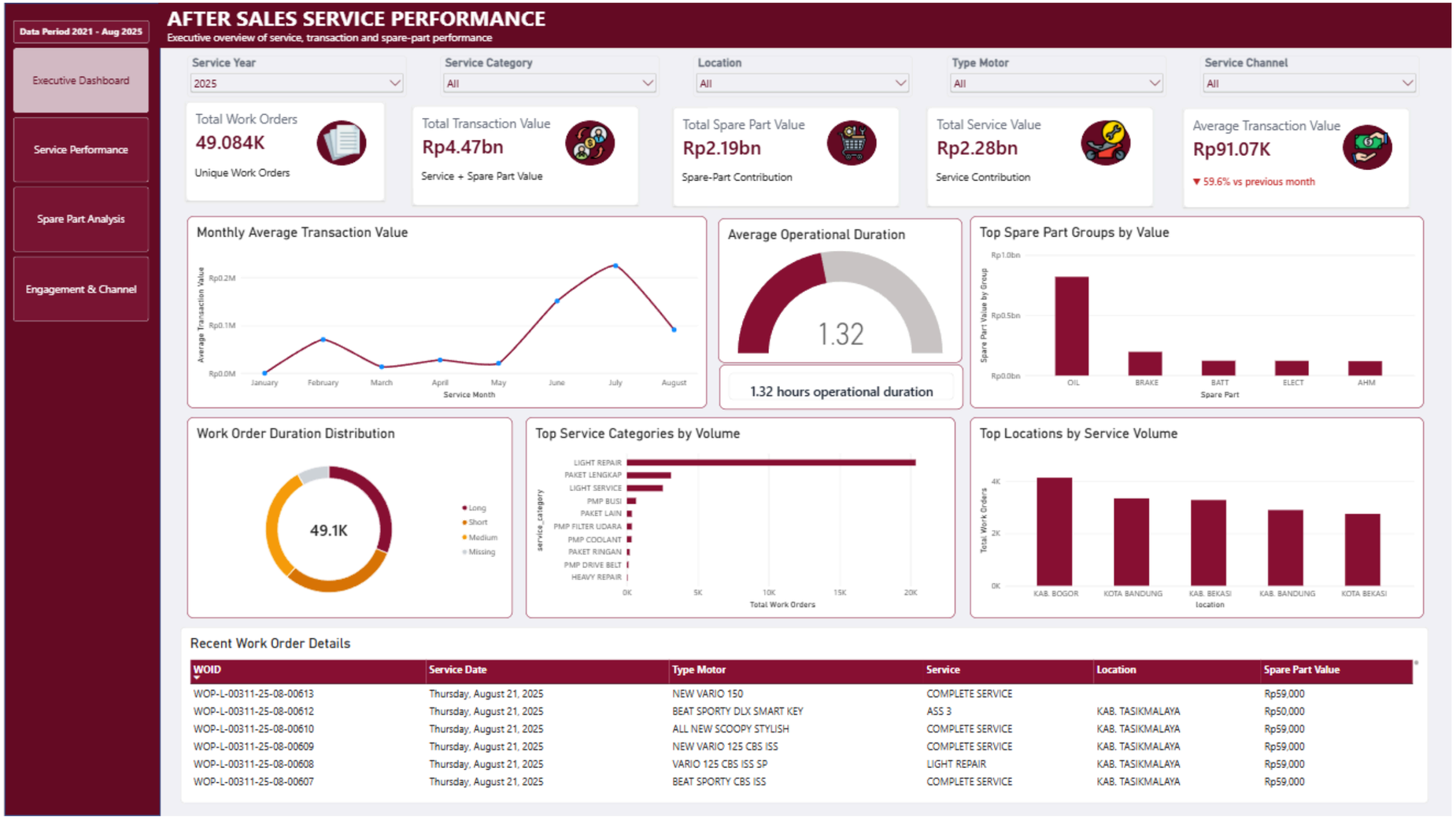
- Compare transaction value and Work Order volume across engagement segments before deciding which campaigns or booking initiatives to expand.



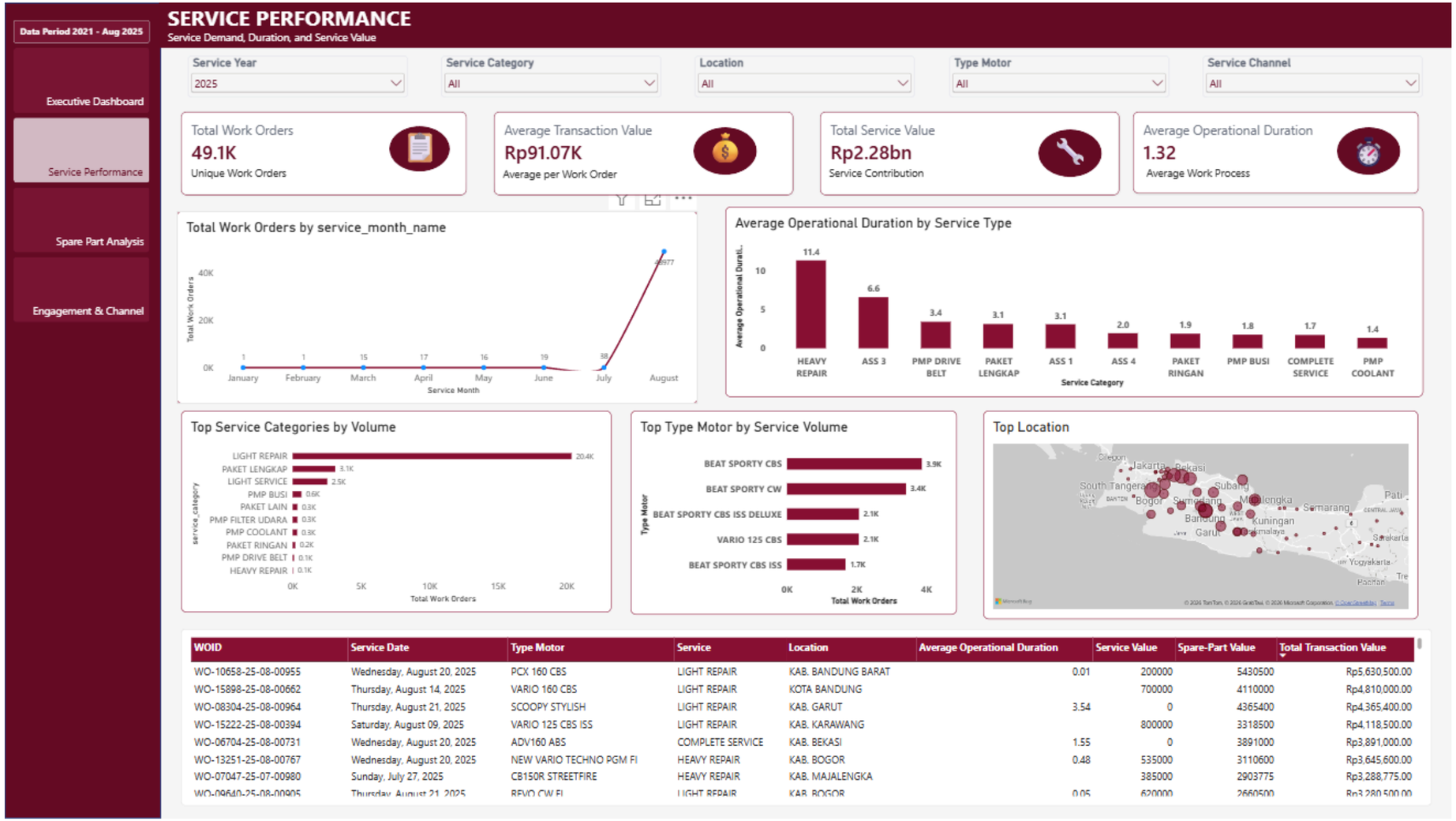
Strengthen Channel Recording and Data Coverage

- Review missing channel information and validate monthly data completeness before assessing channel performance or long-term trends.

7 Dashboard Overview

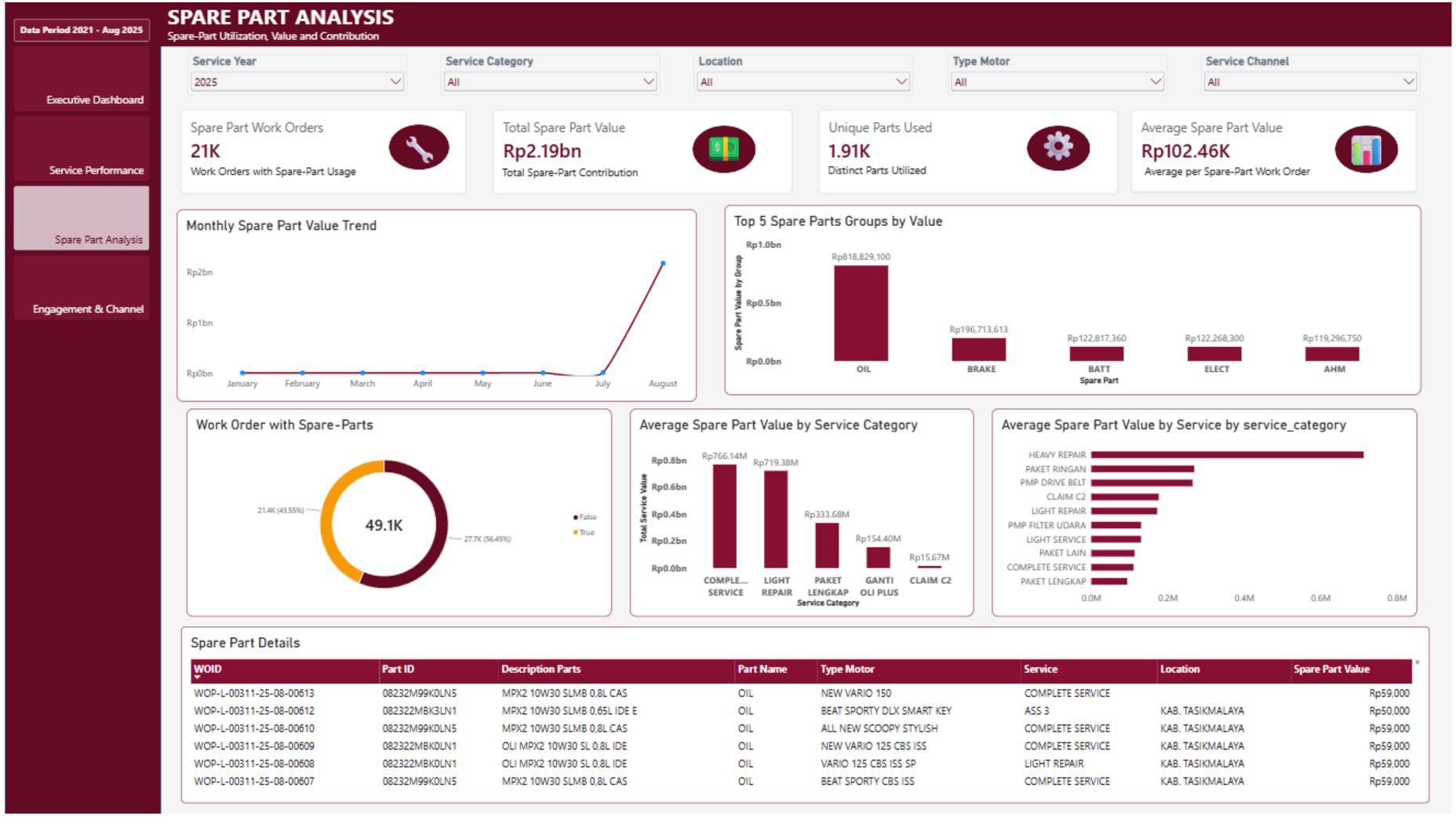


7 Dashboard Overview

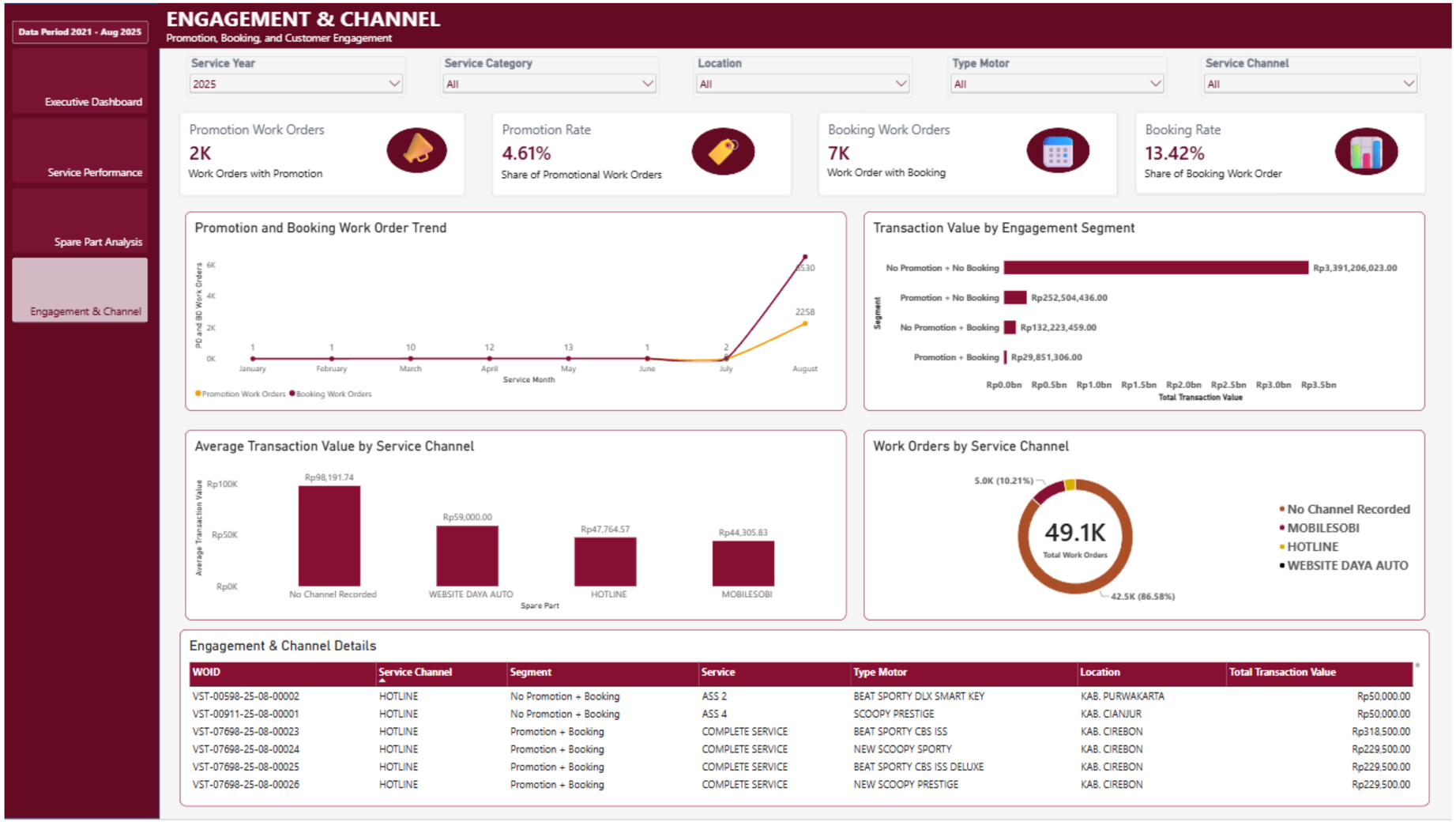


WOID	Service Date	Type Motor	Service	Location	Average Operational Duration	Service Value	Spare-Part Value	Total Transaction Value
WO-10658-25-08-00955	Wednesday, August 20, 2025	PCX 160 CBS	LIGHT REPAIR	KAB. BANDUNG BARAT	0.01	200000	5430500	Rp5,630,500.00
WO-15898-25-08-00662	Thursday, August 14, 2025	VARIO 160 CBS	LIGHT REPAIR	KOTA BANDUNG		700000	4110000	Rp4,810,000.00
WO-08304-25-08-00964	Thursday, August 21, 2025	SCOOPY STYLISH	LIGHT REPAIR	KAB. GARUT	3.54	0	4365400	Rp4,365,400.00
WO-15222-25-08-00394	Saturday, August 09, 2025	VARIO 125 CBS ISS	LIGHT REPAIR	KAB. KARAWANG		800000	3318500	Rp4,118,500.00
WO-06704-25-08-00731	Wednesday, August 20, 2025	ADV160 ABS	COMPLETE SERVICE	KAB. BEKASI	1.55	0	3891000	Rp3,891,000.00
WO-13251-25-08-00767	Wednesday, August 20, 2025	NEW VARIO TECHNO PGM FI	HEAVY REPAIR	KAB. BOGOR	0.48	535000	3110600	Rp3,645,600.00
WO-07047-25-07-00980	Sunday, July 27, 2025	CB150R STREETFIRE	HEAVY REPAIR	KAB. MAJALENGKA		385000	2903775	Rp3,288,775.00
WO-09640-25-08-00905	Thursday, August 21, 2025	REVO CW FI	LIGHT REPAIR	KAB. BOGOR	0.05	620000	2660500	Rp3,280,500.00

7 Dashboard Overview



7 Dashboard Overview



WOID	Service Channel	Segment	Service	Type Motor	Location	Total Transaction Value
VST-00598-25-08-00002	HOTLINE	No Promotion + Booking	ASS 2	BEAT SPORTY DLX SMART KEY	KAB. PURWAKARTA	Rp50,000.00
VST-00911-25-08-00001	HOTLINE	No Promotion + Booking	ASS 4	SCOOPY PRESTIGE	KAB. CIANJUR	Rp50,000.00
VST-07698-25-08-00023	HOTLINE	Promotion + Booking	COMPLETE SERVICE	BEAT SPORTY CBS ISS	KAB. CIREBON	Rp318,500.00
VST-07698-25-08-00024	HOTLINE	Promotion + Booking	COMPLETE SERVICE	NEW SCOOPY SPORTY	KAB. CIREBON	Rp229,500.00
VST-07698-25-08-00025	HOTLINE	Promotion + Booking	COMPLETE SERVICE	BEAT SPORTY CBS ISS DELUXE	KAB. CIREBON	Rp229,500.00
VST-07698-25-08-00026	HOTLINE	Promotion + Booking	COMPLETE SERVICE	NEW SCOOPY PRESTIGE	KAB. CIREBON	Rp229,500.00

Thankyou!



Muhammad Hafizh Rifandi

Data Analyst | Business Intelligence

rifandi.hafizh11@gmail.com

Hafizh Rifandi

Github

+62 851-7518-7050

Jakarta, Indonesia